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Unit 3 - Designing Solutions

PDF Documents

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This PDF document is included in the unit materials.

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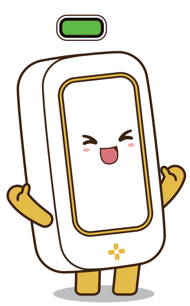
UNIT 3 DESIGNING SOLUTIONS

Algorithm Writing

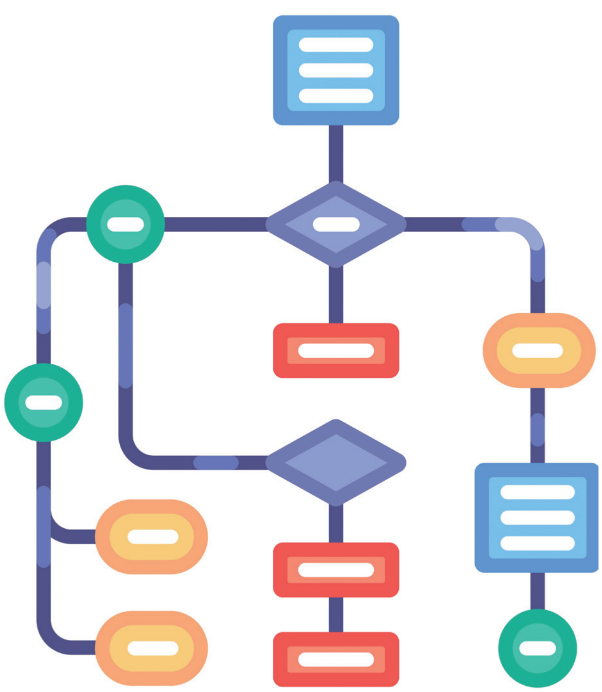
Unit Introduction

In Unit 3, your team will focus on designing the solutions needed to make your smart device work. This involves creating flowcharts that map out the logic of your device, writing the algorithms that will control its actions, and planning the overall system architecture. Your teamwork will be essential as you work together to bring your design to life!



HERE'S WHAT YOU'LL DO:

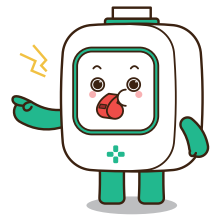
1. Algorithm Writing: Translate the flowcharts into algorithms that will guide the programming of your device.
2. Flowchart Development: Work together to create detailed flowcharts that outline the step-by-step logic your smart device will follow.
3. System Architecture Planning: Design the overall system architecture, including how the MODI modules will interact and communicate.
4. Review and Refine: Continuously review and refine your designs to ensure they are clear, logical, and effective.



Developing Algorithm

TRANSLATING IDEAS TO ALGORITHMS

Use the ideas you developed in Unit 2 as a starting point. For each idea, write a corresponding algorithm. Be detailed and specific, explaining each action and decision your device needs to make.

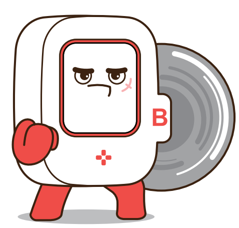


STRUCTURING YOUR ALGORITHMS

Organize your algorithms logically. Break down complex tasks into smaller, manageable steps. Ensure that each part of the algorithm flows smoothly into the next, creating a clear and logical progression.

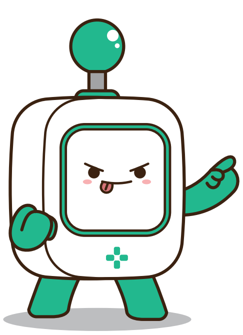
DETAILING THE STEPS

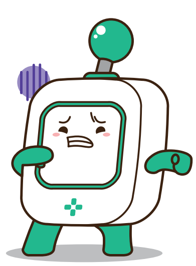
For each step in your algorithm, specify the inputs, processes, and outputs. Clearly define what each MODI module will do at every stage. This detail will help you later when you start creating flowcharts and coding your device.



REVIEWING AND REFINING

Review your algorithms to check for clarity and completeness. Make sure each step is easy to understand and follow. Refine your algorithms as needed to ensure they are as efficient and effective as possible.





Final Thoughts

SMART BEVERAGE VENDING MACHINE

Goal :

To create a smart beverage vending machine that allows users to select a beverage and dispense it automatically.

LOGIC (IDEAS)

1. System Start:
- Power on the vending machine.
- Display a welcome message on the screen: "Welcome! Please select your beverage."

2. Beverage Selection:
- Wait for the user to turn the dial.
- If the dial is turned to position 1, display "Beverage 1 Selected."
- If the dial is turned to position 2, display "Beverage 2 Selected."
- If the dial is turned to position 3, display "Beverage 3 Selected."

3. Confirm Selection:
- Wait for the user to press the button.
- Once the button is pressed, confirm the selected beverage on the display: "Dispensing Beverage X."

4. Dispensing Process:
- Activate Motor A to rotate to the correct position for the selected beverage.
- Start pouring the selected beverage.
- Play pouring sound via the speaker module.

5. Completion:
- When the beverage is fully dispensed, stop Motor A.
- Display "Your Beverage is Ready" on the screen.
- Play a completion sound via the speaker.

6. Reset System:
- Reset the system to the initial state for the next user.
- Display "Welcome! Please select your beverage."

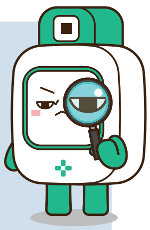
LOGIC (IDEAS)

- User Input: Select beverage with a dial.
- Confirmation: Press a button to confirm the choice.
- Dispensing: Machine dispenses the selected beverage.
- Feedback: Provide feedback through display and sound.

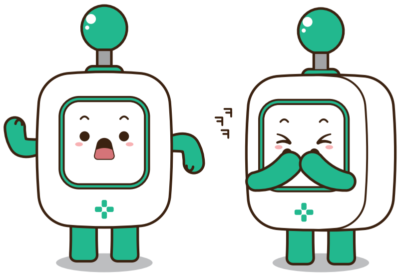
ALGORITHM

- Wait for the user to turn the dial. If in position 1, 2, or 3, display the selected beverage.
- Wait for the button press. Upon press, confirm selection and display "Dispensing Beverage X."
- Rotate Motor A to the correct position, then start pouring the selected beverage.
- Display messages during and after dispensing, and play sounds through the speaker.

TRANSLATION PROCESS:



- **Identify Key Actions:**
  - Break down the idea into distinct actions the machine must perform.
- **Match Actions to Functions:**
  - Determine how each action translates into a function or step in the algorithm.
- **Write the Algorithm:**
  - Translate each function into a clear and precise algorithm step.



Exercise

Now, it's your turn to refine your previous algorithm!

ALGORITHM LOGIC (IDEAS)	TRANSLATION FOR MODI MODULES

(Continue...)

ALGORITHM LOGIC (IDEAS)	TRANSLATION FOR MODI MODULES

Structuring Your Algorithms

Organize your algorithms logically. Break down complex tasks into smaller, manageable steps. Ensure that each part of the algorithm flows smoothly into the next, creating a clear and logical progression.

INSTRUCTIONS

Put Yourself in the MODI Modules' Shoes:

- Imagine you are one of the MODI modules to see if there are any missing steps in your algorithm.

Break Down the Steps:

- Think about how you can break down the steps you wrote before into smaller, more detailed steps.

Refine Your Algorithm:

- Write down the big steps you wrote earlier and break them down into smaller, more detailed steps.

Example

YOUR TRANSLATION

1. System Starts.
2. If the Environment module detects humidity below 60%,
3. The Display module shows the humidity level.
4. Set Motor A to 60°.
5. Maintain the set angle until the humidity level reaches 75%.

REFINED ALGORITHM

1. System Starts.
2. If the Environment module detects humidity below 60%,
3. Shows the humidity level on the Display module and Set Motor A to 60°.
4. Maintain the set angle of Motor A until the humidity level reaches 75%.
5. If the humidity level reaches 75%, set Motor A to the original angle

Exercise

Now, it's your turn to refine your previous algorithm!

YOUR TRANSLATION	TRANSLATION FOR MODI MODULES

Detailing the Steps & Reviewing and Refining

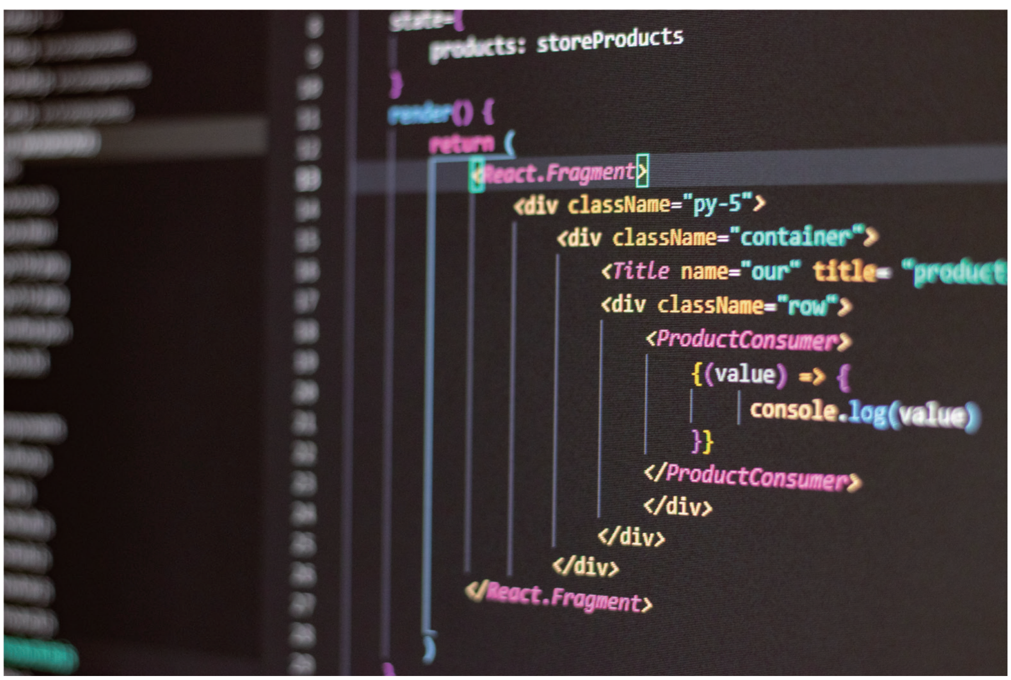
This time, we are going to detail the steps (our last check) and review the entire algorithm to ensure the flow is well-structured.

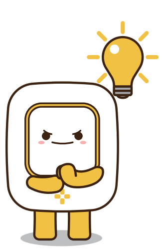
DETAILING THE STEPS

For each step in your algorithm, specify the inputs, processes, and outputs. Clearly define what each MODI module will do at every stage. This detail will help you later when you start creating flowcharts and coding your device.

REVIEWING AND REFINING

Use the ideas you developed in Unit 2 as a starting point. For each idea, write a corresponding algorithm. Be detailed and specific, explaining each action and decision your device needs to make. Look at the example in the previous page, and translate your ideas to algorithm.





Read your refined algorithm you wrote in the previous unit. Is there any missing step? If there isn't, you are ready to draw a flowchart based on the algorithm! Let's move onto the next step, "flowchart"!

Flowchart Development

Unit Introduction

In Unit 4, you will create detailed flowcharts based on the algorithm you wrote in Unit 2 to plan and visualize the logic and steps of your smart device project. Flowcharts are essential tools that help you organize your thoughts, break down complex processes, and ensure that every part of your project works smoothly. This unit will guide you through the process of developing flowcharts for your top three ideas, allowing you to see the big picture and identify any potential issues before you start building.

YOUR ANSWER

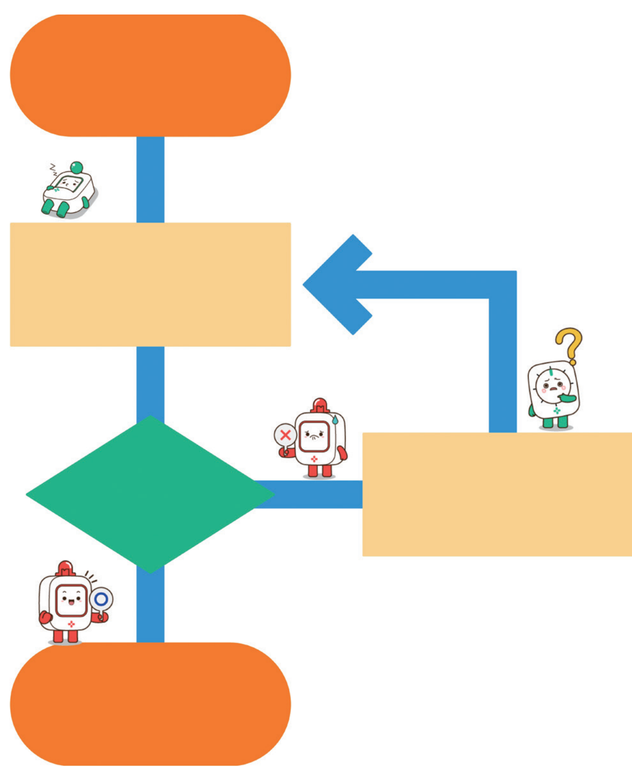
Do you remember the flowcharts you read and sometimes created in the previous chapters? Now, it's time to create your own from the ground up! Exciting!

WHY IS A FLOWCHART IMPORTANT?

Flowcharts are essential tools in planning and developing your smart device project. Here's why they are important:

With flowcharts, you can

- Organize Your Thoughts
- Visualize the Process
- Identify Issues Early
- Improve Communication
- Guide Your Work
- Enhance Problem-Solving



Developing Flowcharts

Creating Flowcharts for Your Ideas

Start by creating flowcharts for the top three ideas you selected in Unit 2. Use symbols like ovals for start and end points, rectangles for actions, diamonds for decisions, and arrows to show the flow of the process. Ensure that each step is clear and logical.

Detailing the Steps

Break down each action in your flowchart into detailed steps. Think about what needs to happen at each stage and what decisions need to be made. This will help you understand the sequence of operations and how different components interact.

Integrating MODI Modules

Incorporate the MODI modules you selected into your flowcharts. Specify which module performs each action and how they work together. This will give you a clear idea of how to program and connect the modules in your project.

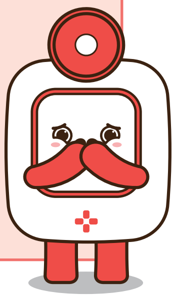
Integrating MODI Modules

Incorporate the MODI modules you selected into your flowcharts. Specify which module performs each action and how they work together. This will give you a clear idea of how to program and connect the modules in your project.

Reviewing and Refining

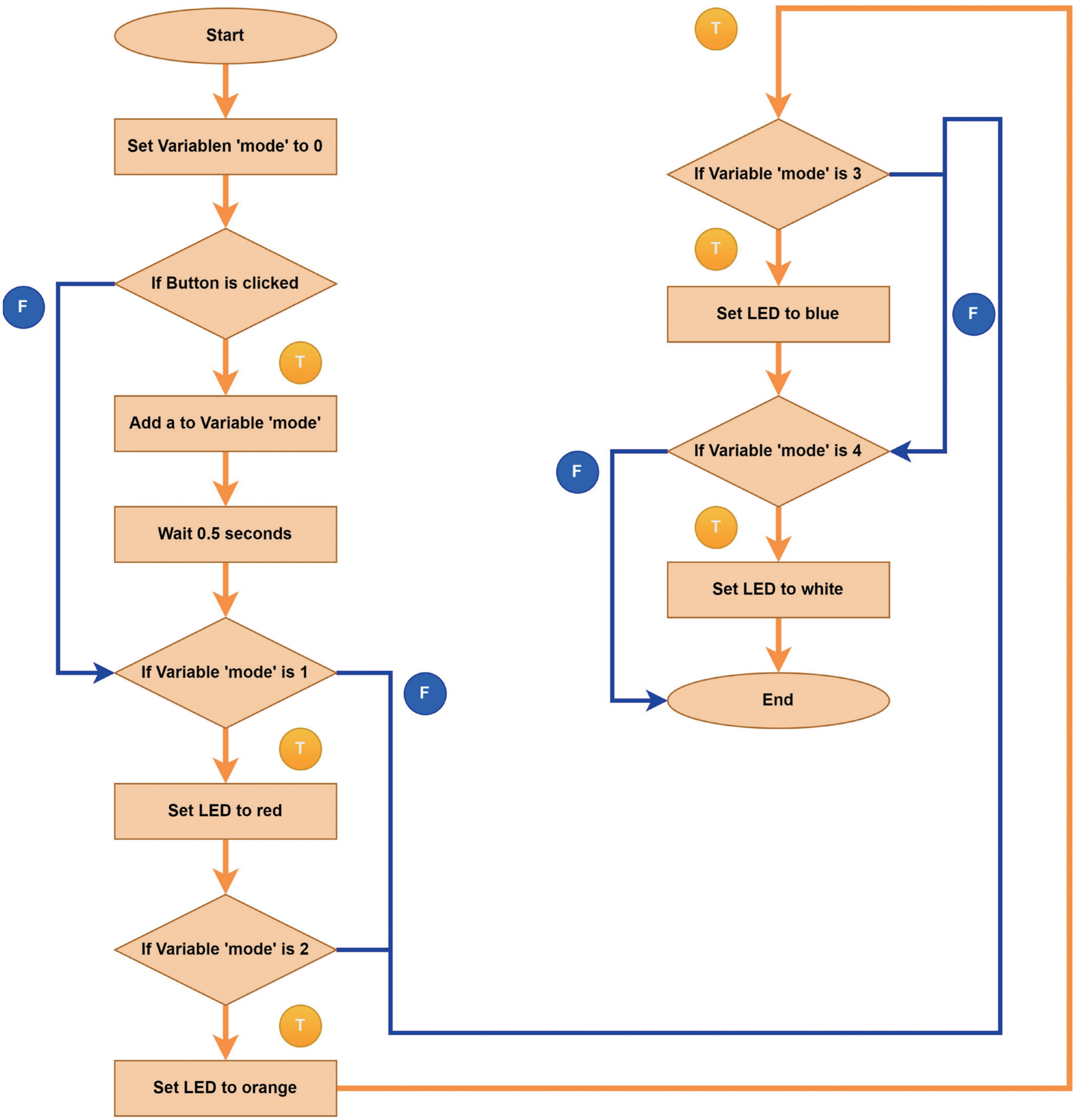
Review your flowcharts to check for any gaps or errors. Make sure that the logic flows smoothly and that all possible scenarios are covered. Refine your flowcharts as needed to ensure they are complete and accurate.

FLOWCHART BRAIN STORMING



Examples of Flowchart

1



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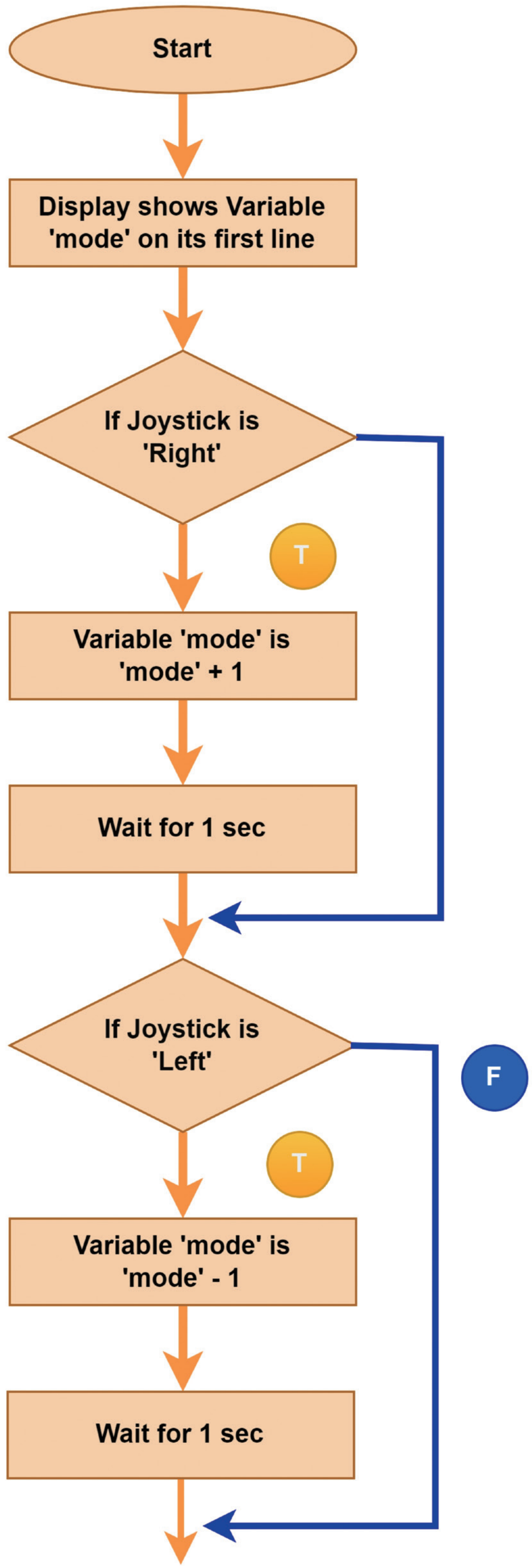
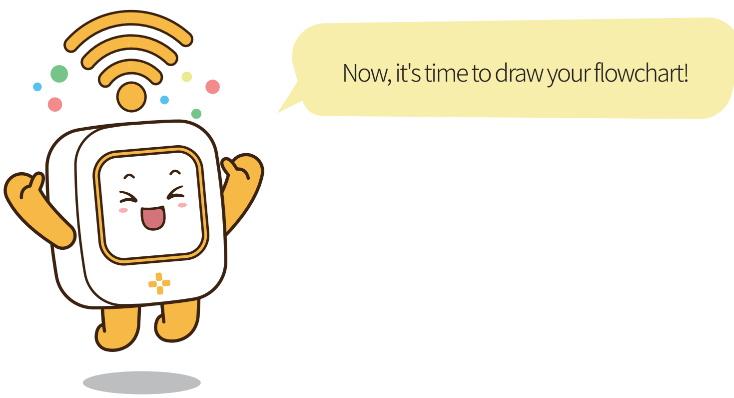
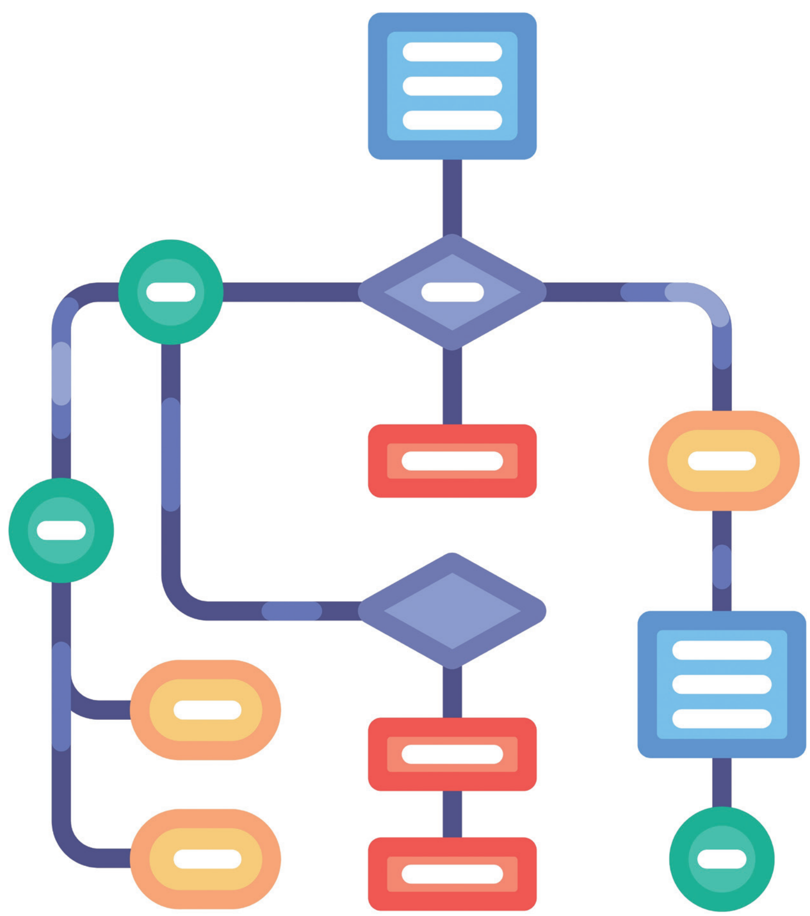

Start
Display shows Variable 'mode' on its first line
If Joystick is 'Right'
T
Variable 'mode' is 'mode' + 1
Wait for 1 sec
If Joystick is 'Left'
F
T
Variable 'mode' is 'mode' - 1
Wait for 1 sec

Drawing Section

Questioning the Details

- What are the key steps and decisions in your project?
- How will the MODI modules be integrated into each step of the process?
- Are there any potential issues or challenges in the flow of your project?
- What improvements can you make to ensure the flowchart is complete and accurate?





YOUR FLOWCHART